

ISAAC McMAHON

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EDUCATION

University of Zürich

PhD in Physics

Jan 2024 – Mar 2027 (Expected)

Zürich, ZH, Switzerland

University of Michigan

MS in Physics

2021 – 2023

Ann Arbor, MI, USA

Carnegie Mellon University

BS in Physics, Astrophysics Track

2017 – 2021

Pittsburgh, PA, USA

EXPERIENCE

Soares-Santos Group (U. of Zürich)

PhD Candidate (PI: Marcelle Soares-Santos)

Jan 2024 – Present

Zürich, ZH, Switzerland

- Led and developed a dark siren cosmology study using the DES galaxy catalog, majorly contributing to the LIGO/Virgo/KAGRA GWTC-5.0 cosmology paper.
- Main developer and contributor of the DESGW gravitational wave counterpart follow-up program with the Dark Energy Camera in LIGO/Virgo/KAGRA fourth observing run. Performed observing, processing, and analysis of images for the purpose of discovering transients possibly associated with gravitational wave events.

LIGO Livingston Observatory (California Inst. of Tech.) Dec 2024 – June 2025

Visiting Fellow

Livingston, LA, USA

- Built, tested, and installed a Seismic Isolation System into the LIGO input laser vacuum chamber during the April 2025 observing break

Soares-Santos Group (U. of Michigan)

PhD Candidate (PI: Marcelle Soares-Santos)

Aug 2021 – Dec 2023

Ann Arbor, MI, USA

- Developed, upgraded, and operated the DESGW image processing pipeline for optical counterparts to GW events for the LIGO/Virgo/KAGRA fourth observing run.

McWilliams Center for Cosmology (Carnegie Mellon U.) May 2020 – Jul 2021

Undergraduate Researcher (PI: Rachel Mandelbaum)

Pittsburgh, PA, USA

- Characterized the completeness of galaxy surveys with Self-Organizing Kohonen Maps

Space Radiation Lab (NASA/Jet Propulsion Lab.)

Undergraduate Research Intern (PI: Mark Wiedenbeck)

Jun 2019 – August 2019

Pasadena, CA, USA

- Modeled solar electron response data in EPI-Hi, an instrument on the Parker Solar Probe, using GEANT4 particle physics simulations

Cold Atom Group (Air Force Research Lab.)
Undergraduate Research Intern (PI: Matthew Squires)

May 2018 – Aug 2018
Albuquerque, NM, USA

- Developed and fabricated novel Magneto-Optical Traps to improve atomic clocks, accelerometers, and other applications, culminating in patent US11467330B1

ACCEPTED PROPOSALS

GW-PRIMO

Principal Investigator

Semester 2026A

CTIO Blanco DECam

- Awarded 2.3 nights for a survey to improve photo-z estimation using medium-band filters for better dark siren precision. NOIRLab Proposal ID 2026A-271131.

DESGW

Co-Investigator

Semesters 2024B-2026B

CTIO Blanco DECam

- Awarded Target-of-Opportunity time to search for EM counterparts of GW events. NOIRLab Proposal IDs 2024B-305328, 2025A-227091, 2025B-485252, 2026B-968899.

SOAR/Goodman ToO for LVK IR1

Co-Investigator

Semester 2026B

CTIO SOAR Goodman

- Awarded Target-of-Opportunity time to classify kilonova candidates from GW EM counterpart searches. NOIRLab Proposal ID 2026B-279715.

AWARDS AND RECOGNITION

CBC Outstanding Contribution

LVK Compact Binary Coalescences Working Group

June 2026

- In recognition of contribution to the fifth Gravitational-wave Transient Catalog

TEACHING

Introduction to Astroparticle Physics and Cosmology

Teaching Assistant

F24

U. of Zürich

Astronomical Techniques

Graduate Student Instructor

F23

U. of Michigan

Introductory Astronomy

Graduate Student Instructor

F21, S22, Su22, F22, S23, Su23

U. of Michigan

Physics II for Engineers

Undergraduate Teaching Assistant

F19, S20, F20, S21

Carnegie Mellon U.

THESIS SUPERVISION

Jan Jonathan Hellwig (BS in Physics, *U. of Zürich*)

Jun 2026

Yavuz Gençel (MS in Physics, *U. of Zürich*)

Dec 2025

OTHER ACTIVITIES

Scientific Organizing Committee

3rd GWCats Meeting

Nov 2026

Toulouse, France

Science Writer

LVK GWTC-5.0 Cosmology Press Release

May 2026

Kitt Peak National Observatory

Support Observer, DESI Survey

Apr 2024

Kitt Peak, AZ, USA

SKILLS

Languages: English (*native*), Mandarin (*advanced*), German (*basic*)

Computing: Python, bash, HPC/slurm/condor, SQL, C++/C#, IDL, HTML/CSS

REFEREED PUBLICATIONS

Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog

McMahon et al. *MNRAS*, 550(1), stag1167 (2026)

– Main contributor, developer, and writer. Wrote all code and ran all analyses.

GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation

LIGO/Virgo/KAGRA Collaborations (2026) pre-print arXiv:2605.27227

– Member of paper writing team and analysis team. Responsible for Data and Discussion sections.

A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph

Zhang, Kokubo, MacBride, **McMahon** et al. (2025) pre-print arXiv:2508.00291

– Coordinated DECam observations, supervised DESGW image processing and candidate vetting.

A broker-integrated algorithm for electromagnetic counterpart searches to gravitational-wave events in O4a and O4b runs

Bommireddy, Forster, **McMahon** et al. *A&A*, 708, A325 (2026)

– Developed and used machine learning code for estimating GW chirp masses.

CONFERENCE PRESENTATIONS

H₀ with GW and the Dark Energy Survey, Swiss Cosmology Days

Jun 2026

Adapting DES for Dark Sirens Analysis, Shedding Light on Dark Sirens

Sep 2025

Improving Dark Siren Measurements of H₀ with GW, APS April Mtg.

Apr 2024

Improving GW Parameter Estimation with Survey Data, DES Collab. Mtg.

Oct 2023

SEMINARS

<i>LIGO: Listening for Gravitational Waves</i> , St Charles Parish Rotary Club	Jun 2025
<i>A Short Guide to Multi-Messenger Cosmology</i> , LSU Astrophysics Seminar	Sep 2023
<i>A Short Guide to GW Cosmology</i> , UM Physics Graduate Student Symposium	Jul 2023

POSTERS

<i>Dark Sirens with DES Y6 and GWTC-4.0</i> , Swiss Physics Soc. Annual Mtg.	Aug 2026
<i>Dark Sirens with the DES Y3 Galaxy Catalogue</i> , LVK Collab. Mtg. • Poster Prize Winner	Sep 2024
<i>Visual Study of Incompleteness in Photo-z Estimation Using SOMs</i> , AAS 237	Jan 2021